

Jiacheng (J.C.) Wen

Math + CS @ UC Berkeley | Machine Learning & Multi-Agent Systems

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Education

University of California, Berkeley

Berkeley, CA

B.A. Mathematics; B.A. Computer Science GPA: 4.0/4.0

Aug 2026 – Expected May 2030

- Relevant Coursework: Structure and Interpretation of Computer Programs, Data Structures and Algorithms, Foundations of Data Science, Discrete Mathematics

Skyline College

San Bruno, CA

A.A. Mathematics; A.A. Computer Science GPA: 4.0/4.0

Jun 2024 – May 2026

- Relevant Coursework: Intro Object-Oriented Programming (C++), Discrete Math for CS, Probability for CS, Computer Networking, Calculus I/II/III, Linear Algebra, Differential Equations

Research Experience

Computer Vision Research

Berkeley, CA

University of California, Berkeley – Undergraduate Research Assistant

Sep 2026 – Present

- Implemented, tested, and documented blink-detection algorithms in Python, extending an existing lab codebase
- Built a hand-annotated ground-truth dataset by labeling blink onset/offset events across eye-tracking video recordings
- Maintained the codebase under Git/GitHub version control with unit tests, example data, and usage documentation
- Produced publication-quality figures and written summaries of methods and results for the research team

Machine Learning & Multi-Agent Systems Research

Pittsburgh, PA

Carnegie Mellon University

Jun 2025 – Dec 2025

- Collaborated with a CMU professor and undergraduate research team to design multi-agent reinforcement learning experiments modeling competitive agent behavior
- Built Python simulations training two competing agents via self-play with reward-and-penalty mechanics in a fighting-game-style environment
- Modeled agent interaction dynamics mathematically to evaluate policy convergence and strategy emergence

Projects

Multi-Agent Reinforcement Learning Simulation | *Python, NumPy, MARL, Self-Play*

2025

- Built a Python simulation of two competing AI agents trained via self-play with reward-and-penalty mechanics in a fighting-game-style environment
- Applied policy optimization and algorithm design to iterate on agent strategy

Malicious URL & Phishing Detector | *Python, pandas, scikit-learn, Streamlit*

2026

- Engineered lexical and structural features (URL length, subdomain count, suspicious TLDs/keywords) from 650k+ labeled URLs to detect phishing links
- Trained a Random Forest classifier with balanced class weights to prioritize recall on malicious URLs, achieving 83% accuracy (0.80 recall / 0.72 precision on the malicious class)
- Deployed an interactive Streamlit app for real-time URL classification

SN5 Telemetry Site | *Python, FastAPI, pandas, SvelteKit*

2026

- Built a full-stack app to upload and visualize racecar CAN-bus telemetry, backed by a FastAPI REST API and SQLite
- Implemented z-score-based anomaly detection to automatically flag statistically significant deviations in sensor signals

Scheme Interpreter | *Python, Scheme*

2026

- Developed a Read-Eval-Print Loop (REPL) Scheme interpreter in Python
- Implemented syntactic parsing to convert raw code strings into nested AST data structures

Additional

Awards: AMC12 Top 5%, AIME Qualifier, USACO Silver

Technical Skills: Python, PyTorch, NumPy, pandas, scikit-learn, Git/GitHub, FastAPI, Java, C++, JavaScript, HTML/CSS, SQL, Scheme, Jupyter Notebook

Languages: Chinese (Native), English (Fluent)